



SAFETY DATA SHEET

SOLITAIRE

SECTION 1. PRODUCT AND COMPANY IDENTIFICATION

Product name : SOLITAIRE

Other means of identification : Not applicable

Recommended use : Manual Warewashing Detergent

Restrictions on use : Reserved for industrial and professional use.

Product dilution information : 0.03 % - 0.08 %

Company :

Chile ECOLAB S.A., Avenida Pedro de Valdivia 3801, Santiago, Chile, (2)-22413300, Telephone: (2)-22381603, Telephone: SAC: 600 241 6600, sac.chile@ecolab.com

Ecolab Argentina S.R.L., Street 11 N° 978, Industrial Park Pilar - Pilar, Buenos Aires, Zip Code: B1629MXA, Telephone: 0230 467 1000

Ecolab Colombia S.A., Avenida Carrera 60 No. 22 – 75 Interior 7., Bogotá, Colômbia, Telephone: 01 8000-120020

ECOLAB Costa Rica SRL, BES Free Zone, El Coyoil 7 km west Juan Santamaria Airport, Alajuela, Costa Rica, ZIP Code: Apdo109-4003

Ecolab Mexico, Avenida Industriales 28, Fraccionamiento Industrial - Cuamatla, Cuautitlán Izcalli, Mexico State, Zip Code: 54730, Telephone: (0155) 5864-1800

Quimiproduktos S. de R.L. de C.V., Carretera Miguel Alemán, 990, Colonia La Encarnación - Apodaca, Nuevo León, Mexico, Zip Code: 66633, Telephone: (0181) 47395900

Peru, Calle Santa Isabel # 326 Miraflores. Lima - Perú, Telephone: 511 7176200

Emergency telephone :

Argentina: Ciquime: 0800-222-2933/ 011 4613-1100, National Poison Control Center: 0800 333 0160

Chile: Poison Information Centre CITUC: (2) 3800 -2635

Colômbia: CISPROMQUIM® (24 hour service) Phone: 2886012 (Bogotá), Telephone: 018000916012 (Colombia), Telephone: 08001005012 (Venezuela), Telephone: 1800-59-3005 (Ecuador: solo Quito, La sierra, Centro and norte), Telephone: Ecolab USA and Canadá 1-651-222-5352

Costa Rica: For medical emergency information, call 911 and (506) 2436-3900 in Costa Rica. National Poison Control Center in Costa Rica: (506)2223-1028

Mexico: Telephone: 01 800 837 2998 Please request assistance in spanish language

Mexico State: Telephone: 01 800 837 2998

Peru: Telephone: 1-651-222-5352 Please request assistance in spanish language

Issuing date : 15.02.2016

SECTION 2. HAZARDS IDENTIFICATION

Product AS SOLD

Physical and chemical hazards : Not flammable or combustible.

Potential Health Effects

Eyes : Causes eye irritation.

Skin : Causes skin irritation.

Ingestion : May be harmful if swallowed.

Inhalation : Health injuries are not known or expected under normal use.

SAFETY DATA SHEET

SOLITAIRE

Chronic Exposure : Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Physical and chemical hazards : Not flammable or combustible.

Potential Health Effects

Eyes : Health injuries are not known or expected under normal use.

Skin : Health injuries are not known or expected under normal use.

Ingestion : Health injuries are not known or expected under normal use.

Inhalation : Health injuries are not known or expected under normal use.

Chronic Exposure : Health injuries are not known or expected under normal use.

SECTION 3. COMPOSITION/INFORMATION ON INGREDIENTS

Product AS SOLD

Pure substance/mixture : Mixture

Chemical name	CAS-No.	Concentration (%)
sodium dodecylbenzene sulfonate	25155-30-0	30 - 60
Polyethylene Glycol	25322-68-3	10 - 30
dodecanamide, n-(2-hydroxyethyl)-	142-78-9	10 - 30
Sodium poly(oxyethylene) dodecyl ether sulfate	68585-34-2	5 - 10
Alkylamineoxides	68155-09-9	5 - 10
acetic acid, sodium salt	127-09-3	1 - 5
ethanol	64-17-5	1 - 5

Product AT USE DILUTION

No hazardous ingredients

SECTION 4. FIRST AID MEASURES

Product AS SOLD

In case of eye contact : Rinse with plenty of water.

In case of skin contact : Wash off immediately with plenty of water for at least 15 minutes. Use a mild soap if available. Get medical attention if irritation develops and persists.

If swallowed : Rinse mouth. Get medical attention if symptoms occur.

If inhaled : Get medical attention if symptoms occur.

Protection of first-aiders : If potential for exposure exists refer to Section 8 for specific personal protective equipment.

Notes to physician : Treat symptomatically.

Most important symptoms and effects, both acute and delayed : See Section 11 for more detailed information on health effects and symptoms.

SAFETY DATA SHEET

SOLITAIRE

Product AT USE DILUTION

In case of eye contact	: Rinse with plenty of water.
In case of skin contact	: Rinse with plenty of water.
If swallowed	: Rinse mouth. Get medical attention if symptoms occur.
If inhaled	: Get medical attention if symptoms occur.

SECTION 5. FIRE-FIGHTING MEASURES

Product AS SOLD

Suitable extinguishing media	: Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.
Unsuitable extinguishing media	: None known.
Specific hazards during fire fighting	: Not flammable or combustible.
Hazardous combustion products	: Decomposition products may include the following materials: Carbon oxides Nitrogen oxides (NOx) Sulfur oxides Oxides of phosphorus
Special protective equipment for fire-fighters	: Use personal protective equipment.
Specific extinguishing methods	: Collect contaminated fire extinguishing water separately. This must not be discharged into drains. Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.

SECTION 6. ACCIDENTAL RELEASE MEASURES

Product AS SOLD

Personal precautions, protective equipment and emergency procedures	: Ensure clean-up is conducted by trained personnel only. Refer to protective measures listed.
Environmental precautions	: Do not allow contact with soil, surface or ground water.
Methods and materials for containment and cleaning up	: Sweep up and shovel into suitable containers for disposal.

Product AT USE DILUTION

Personal precautions, protective equipment and emergency procedures	: Refer to protective measures listed.
Environmental precautions	: No special environmental precautions required.
Methods and materials for containment and cleaning up	: Stop leak if safe to do so. Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13). For large spills, dike

SAFETY DATA SHEET

SOLITAIRE

spilled material or otherwise contain material to ensure runoff does not reach a waterway. Do not flush into surface water or sanitary sewer system.

SECTION 7. HANDLING AND STORAGE

Product AS SOLD

Advice on safe handling : Do not ingest. Avoid contact with skin and eyes. Wash hands thoroughly after handling.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

Storage temperature : 0 °C to 50 °C

Product AT USE DILUTION

Advice on safe handling : Wash hands after handling. For personal protection see section 8.

Conditions for safe storage : Keep out of reach of children. Store in suitable labeled containers.

SECTION 8. EXPOSURE CONTROLS/PERSONAL PROTECTION

Product AS SOLD

Ingredients with workplace control parameters

Ingredients	CAS-No.	Form of exposure	Permissible concentration	Basis
Polyethylene Glycol	25322-68-3	TWA (aerosol)	10 mg/m ³	US WEEL
ethanol	64-17-5	CMP	1,000 ppm	AR OEL
ethanol	64-17-5	LT	780 ppm 1,480 mg/m ³	BR OEL
ethanol	64-17-5	LPP	875 ppm 1,645 mg/m ³	CL OEL
ethanol	64-17-5	TWA	1,000 ppm 1,884 mg/m ³	PE OEL
ethanol	64-17-5	CAP	1,000 ppm	VE OEL
ethanol	64-17-5	CPT	1,000 ppm 1,900 mg/m ³	PA OEL
ethanol	64-17-5	LMPE-PPT	1,000 ppm 1,900 mg/m ³	MX OEL
ethanol	64-17-5	TWA	1,000 ppm 1,900 mg/m ³	NIOSH REL
		TWA	1,000 ppm 1,900 mg/m ³	OSHA Z1

Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

Eye protection : Due to the form and packaging of the product, no protective equipment is needed under normal use conditions.

Hand protection : Due to the form and packaging of the product, no protective equipment is needed under normal use conditions.

Skin protection : No special protective equipment required.

SAFETY DATA SHEET

SOLITAIRE

- Respiratory protection : When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.
- Hygiene measures : Handle in accordance with good industrial hygiene and safety practice. Wash face, hands and any exposed skin thoroughly after handling.

Product AT USE DILUTION

- Engineering measures : Good general ventilation should be sufficient to control worker exposure to airborne contaminants.

Personal protective equipment

- Eye protection : No special protective equipment required.
- Hand protection : No special protective equipment required.
- Skin protection : No special protective equipment required.
- Respiratory protection : No personal respiratory protective equipment normally required.

SECTION 9. PHYSICAL AND CHEMICAL PROPERTIES

	Product AS SOLD	Product AT USE DILUTION
Appearance	: Cast Solid	liquid
Color	: opaque, light pink	colorless
Odor	: citrus	citrus
pH	: 7.0 - 9.0, 1 %	7.0 - 8.0
Flash point	: Not applicable	
Odor Threshold	: No data available	
Melting point/freezing point	: No data available	
Initial boiling point and boiling range	: No data available	
Evaporation rate	: No data available	
Flammability (solid, gas)	: No data available	
Upper explosion limit	: No data available	
Lower explosion limit	: No data available	
Vapor pressure	: No data available	
Relative vapor density	: No data available	
Relative density	: No data available	
Water solubility	: insoluble	
Solubility in other solvents	: No data available	
Partition coefficient: n-octanol/water	: No data available	
Autoignition temperature	: No data available	
Thermal decomposition	: No data available	
Viscosity, kinematic	: No data available	
Explosive properties	: No data available	

SAFETY DATA SHEET

SOLITAIRE

Oxidizing properties : No data available
Molecular weight : No data available
VOC : No data available

SECTION 10. STABILITY AND REACTIVITY

Product AS SOLD

Chemical stability : Stable under normal conditions.
Possibility of hazardous reactions : No dangerous reaction known under conditions of normal use.
Conditions to avoid : None known.
Incompatible materials : None known.
Hazardous decomposition products : Decomposition products may include the following materials:
Carbon oxides
Nitrogen oxides (NOx)
Sulfur oxides
Oxides of phosphorus

SECTION 11. TOXICOLOGICAL INFORMATION

Information on likely routes of exposure : Eye contact, Skin contact

Potential Health Effects

Product AS SOLD

Eyes : Causes eye irritation.
Skin : Causes skin irritation.
Ingestion : May be harmful if swallowed.
Inhalation : Health injuries are not known or expected under normal use.
Chronic Exposure : Health injuries are not known or expected under normal use.

Product AT USE DILUTION

Eyes : Health injuries are not known or expected under normal use.
Skin : Health injuries are not known or expected under normal use.
Ingestion : Health injuries are not known or expected under normal use.
Inhalation : Health injuries are not known or expected under normal use.
Chronic Exposure : Health injuries are not known or expected under normal use.

Experience with human exposure

Product AS SOLD

Eye contact : Redness, Irritation
Skin contact : Redness, Irritation

SAFETY DATA SHEET

SOLITAIRE

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Product AT USE DILUTION

Eye contact : No symptoms known or expected.

Skin contact : No symptoms known or expected.

Ingestion : No symptoms known or expected.

Inhalation : No symptoms known or expected.

Toxicity

Product AS SOLD

Acute oral toxicity : No data available

Acute inhalation toxicity : No data available

Acute dermal toxicity : No data available

Skin corrosion/irritation : No data available

Serious eye damage/eye irritation : Mild eye irritation

Respiratory or skin sensitization : No data available

Carcinogenicity : No data available

Reproductive effects : No data available

Germ cell mutagenicity : No data available

Teratogenicity : No data available

STOT-single exposure : No data available

STOT-repeated exposure : No data available

Aspiration toxicity : No data available

Ingredients

Acute oral toxicity : sodium dodecylbenzene sulfonate
LD50 Rat: 1,086 mg/kg

Polyethylene Glycol
LD50 Rat: 28,000 mg/kg

dodecanamide, n-(2-hydroxyethyl)-
LD50 Rat: > 2,000 mg/kg

Sodium poly(oxyethylene) dodecyl ether sulfate
LD50 Rat: > 2,500 mg/kg

acetic acid, sodium salt
LD50 Rat: 3,530 mg/kg

ethanol
LD50 Rat: 10,470 mg/kg

Ingredients

SAFETY DATA SHEET

SOLITAIRE

Acute inhalation toxicity : acetic acid, sodium salt
4 h LC50 Rat: > 5.6 mg/l

ethanol
4 h LC50 Rat: 117 mg/l

Ingredients

Acute dermal toxicity : Polyethylene Glycol
LD50 Rabbit: 20,000 mg/kg

Sodium poly(oxyethylene) dodecyl ether sulfate
LD50 Rat: > 2,000 mg/kg

acetic acid, sodium salt
LD50 Rabbit: > 20,000 mg/kg

ethanol
LD50 Rabbit: > 15,800 mg/kg

SECTION 12. ECOLOGICAL INFORMATION

Product AS SOLD

Ecotoxicity

Environmental Effects : Toxic to aquatic life.

Product

Toxicity to fish : No data available

Toxicity to daphnia and other aquatic invertebrates : No data available

Toxicity to algae : No data available

Ingredients

Toxicity to fish : sodium dodecylbenzene sulfonate
96 h LC50: 3.2 mg/l

Polyethylene Glycol
96 h LC50 Fish: > 1,000 mg/l

dodecanamide, n-(2-hydroxyethyl)-
96 h LC50 Fish: > 100 mg/l

Sodium poly(oxyethylene) dodecyl ether sulfate
96 h LC50 Fish: 28 mg/l

acetic acid, sodium salt
96 h LC50 Fish: > 100 mg/l

ethanol
96 h LC50 Pimephales promelas (fathead minnow): > 100 mg/l

Persistence and degradability

No data available

Bioaccumulative potential

No data available

SAFETY DATA SHEET

SOLITAIRE

Mobility in soil

No data available

Other adverse effects

No data available

SECTION 13. DISPOSAL CONSIDERATIONS

Product AS SOLD

Disposal methods : The product should not be allowed to enter drains, water courses or the soil. Where possible recycling is preferred to disposal or incineration. If recycling is not practicable, dispose of in compliance with local regulations. Dispose of wastes in an approved waste disposal facility.

Disposal considerations : Dispose of as unused product. Empty containers should be taken to an approved waste handling site for recycling or disposal. Do not re-use empty containers. Dispose of in accordance with local, state, and federal regulations.

Product AT USE DILUTION

Disposal methods : Diluted product can be flushed to sanitary sewer.

Disposal considerations : Dispose of in accordance with local, state, and federal regulations.

SECTION 14. TRANSPORT INFORMATION

Product AS SOLD

The shipper/consignor/sender is responsible to ensure that the packaging, labeling, and markings are in compliance with the selected mode of transport.

Land transport

Not dangerous goods

Sea transport (IMDG/IMO)

Not dangerous goods

Product AT USE DILUTION

Not intended for transport.

SECTION 15. REGULATORY INFORMATION

For Regulatory information please contact your Ecolab sales representative.

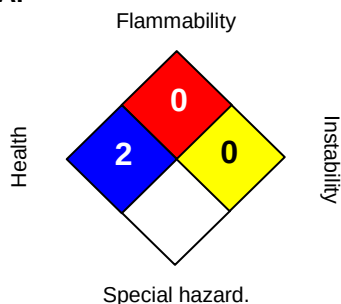
SECTION 16. OTHER INFORMATION

Product AS SOLD

SAFETY DATA SHEET

SOLITAIRE

NFPA:



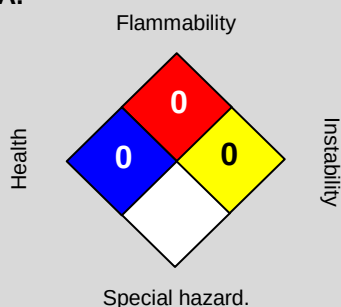
HMIS III:

HEALTH	2
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Product AT USE DILUTION

NFPA:



HMIS III:

HEALTH	0
FLAMMABILITY	0
PHYSICAL HAZARD	0

0 = not significant, 1 = Slight,
2 = Moderate, 3 = High
4 = Extreme, * = Chronic

Issuing date : 15.02.2016
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Prepared by : Regulatory Affairs

REVISED INFORMATION: Significant changes to regulatory or health information for this revision is indicated by a bar in the left-hand margin of the MSDS.

The information provided in this Material Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text.